

Photosynthesis is the process plants use to turn light into chemical energy. It takes place in the chloroplasts, which contain the green pigment chlorophyll. Chlorophyll absorbs red and blue light and reflects green light. The light reactions split water and release oxygen as a by-product. The Calvin cycle uses carbon dioxide to build glucose molecules. Mitochondria later break glucose down during cellular respiration. Cellular respiration releases energy that the cell stores as ATP.